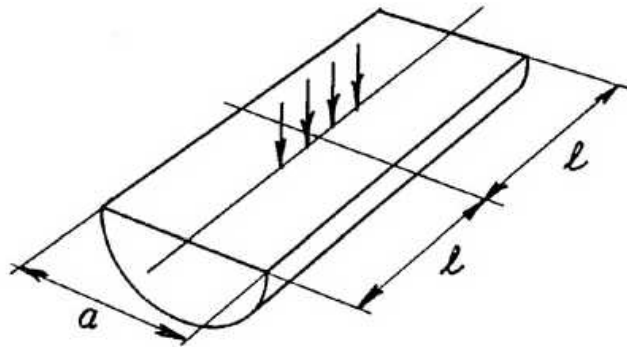


G)

BAR OF SEMI-CIRCULAR CROSS SECTION

The geometric factor for an infinitely long bar of semi-circular cross section is included in (d):

$$\varrho = G \frac{V}{s}, \quad G = \frac{2\pi s}{F}$$

where $F = F(\frac{a}{s})$ is shown at page 33.

The value of F for $a = 10 \cdot s$ is $F(\frac{a}{s} = 10) = 1,038$, or

$$G(\frac{a}{s} = 10) = 0,963 \cdot 2\pi s.$$

When $a \geq 10s$, then $0,963 \cdot 2\pi s \leq G \leq 2\pi s$, where $2\pi s$ is the geometric factor for a semi-infinite volume.

The deviation of G from $G = \frac{2\pi s}{F}$ for a bar of finite

length $2l$ can be estimated from the special case of a box-shaped bar treated by Hansen (d), see p. 31, as the rectangular shape with $h = \frac{1}{2} \cdot a$ is not very different in this context from a semi-circular cross section with diameter a .

So, on the basis of (d) we estimate that when

$$2l \geq 3s + \frac{1}{2}a, \quad \text{then } 0,98 \frac{2\pi s}{F} \leq G \leq \frac{2\pi s}{F}.$$